MSCA Digitial Finance

"Multi-criteria decision making in sustainable finance", Kaiserslautern, 26.-28.10.2026

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Lecture 5: Optimal Investment with Digital Assets and Worst-Case Portfolio Optimization

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Based on joint work with:

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- Olaf Menkens (Dublin City University)
- Lukas Müller (TU Kaiserslautern)
- Frank Seifried (TU Kaiserslautern)
- Mogens Steffensen (Univ. Kopenhagen)
- Paul Wilmott

A preliminary remark – Price movement of digital assets:

- Digital assets are now well-established (Bitcoin, Ethereum, …)
- Price processes include jumps (see e.f. Saev et al. (2024))
- Portfolio optimization with jump-diffusion processes is necessary

Repetition: The Portfolio Optimization Problem

Portfolio-Optimization:

Decide how many shares of which security to hold at what point in time to **maximize the investor's expected utility** from **final wealth at time T** (and from **consumption on [0,T]**).

Part 0: Optimal portfolios with jump-diffusion prices

The Merton problem

Consider a Black-Scholes-type market with a money market account and a stock via

$$dB(t) = B(t)rdt \qquad B(0) = 1, \quad \text{“MMA”}$$

$$dS(t) = S(t-)\big((b - \lambda\kappa)dt + \sigma dW(t) + (Y(t) - 1)dN(t)\big) \quad S(0) = s, \quad \text{“Stock”},$$

with Y(t) being log-normally distributed with $\mathrm{E}(Y(t) - 1) = \kappa$.

Goal: Find a portfolio process $\pi^*(.)$ (**fraction of wealth invested in the stock**) solving

$$v_0(0,x) := \sup_{\pi(.)\in A(x)} E\left(ln\left(X^{\pi}(T)\right)\right) \qquad v_0(t,x) := \sup_{\pi(.)\in A(t,x)} E^{t,x}\left(ln\left(X^{\pi}(T)\right)\right)$$

with wealth process given by

$$\mathrm{dX^{\pi}(t)} = \mathrm{X^{\pi}(t-)}[(\mathrm{r} + \pi(\mathrm{t})(\mathrm{b} - \lambda\kappa))\mathrm{dt} + \pi(\mathrm{t})\sigma\mathrm{dW(t)} + \pi(\mathrm{t}-)(Y(t) - 1))dN(t)],$$

$$\mathrm{X^{\pi}(0)} = \mathrm{x}$$

Solving the portfolio problem with the help of the Itô-formula for jump-diffusion processes

Note that we have

$$\ln\left(X^{\pi}(t)\right) = \ln(x) + \int_0^t \left(r + \pi(s-)(b - \lambda\kappa) - \frac{1}{2}(\pi-)^2\sigma^2\right) ds$$

$$+ \int_0^t \pi(s-)\sigma dW(s) + \int_0^t ln(1 + \pi(s-)(Y(s) - 1))dN(s)$$

Taking expectations yields (assuming that the portfolio process is square integrable), we obtain

$$E(ln(X^{\pi}(t) = ln(x)+$$

$$+ \int_0^t E\left(\left(r + \pi(s-)(b - \lambda\kappa) - \frac{1}{2}(\pi-)^2\sigma^2\right) + \lambda(ln(1 + \pi(s-)(Y(s) - 1)))\right) ds$$

Hence, we have to solve this interior optimization numerically by calculating the second expectation.

More precisely:

By taking the derivative with respect to π, we arrive at the necessary condition of

$$0 = \mathrm{b} - \lambda\kappa - \pi\sigma^2 + \lambda E\left(\frac{Y-1}{1+\pi(Y-1)}\right)$$

As we know the density of Y, we can calculate the expectation for every π. Checking the form of the second order condition shows that the obtained π from the first order condition is the optimal portfolio.

Remark:

i) For the general power utility, we need to set up a generalization of the HJB-equation in the diffusion case which now turns to an integro-partial-differential equation.

ii) In the case of only one possible jump height y, one obtains the quadratic equation

$$0 = \mathrm{b} - \lambda\kappa - \pi\sigma^2 + \lambda\frac{y-1}{1+\pi(y-1)}$$

and has to find the appropriate extremum of it.

iii) Another interesting portfolio problem might be one that includes maximizing the utility from the final wealth after having transferred the wealth to conventional assets.

To present an alternative for working with assets with possible jumps – and thus also with digital assets – is to come up with an alternative jump modelling. This will lead to our second ingredient of this lecture.

Worst-case portfolio optimization – A second preliminary remark:

- Standard portfolio optimization problems/stochastic control problems are based on distributional assumptions for the risky processes entering them ("**Risk**")

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Three versions of Knightian uncertainty in my lecture:

- ❖**Stock price crash:** An event when you loose an a prioi unknown fraction of your wealth invested in all stocks
- ❖**Ambiguity:** There is no perfect knowledge about market coefficients, only that they are in a bounded region
- ❖**A Defaultable reinsurer:** We consider an insurance application motivated by the Swiss solveny test

Part I: The Continous-Time Portfolio Optimization Problem

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$$v_0(t,x) := \sup_{\pi(.)\in A(t,x)} E^{t,x}\Big(U\big(X^\pi(T)\big)\Big)$$

with wealth process

$$dX^\pi(t) = X^\pi(t)\Big[\big(r+\pi(t)\lambda\big)dt+\pi(t)\sigma dW(t)\Big], \quad X^\pi(0)=x,$$

$U(x)$ a utility function (i.e. twice differentiable, concave, strictly increasing)

Solution methods: Martingale methods, convex duality, HJB-equation, …

Some explicitly solved examples:

a) Log- and power utility: $U(x)=\frac{1}{1-\gamma}x^{1-\gamma}$, $1\neq\gamma>0$, $U(x)=\ln(x)$, $("\gamma=1")$

(1) $\pi^*(t)=\frac{\lambda}{\gamma\sigma^2}$ **Constant fraction of wealth in stock**

b) Exponential utility: $U(x)=1-e^{-\gamma x}$, $\gamma>0$, $r=0$

(2) $\pi^*(t)X^{\pi}(t)=\frac{\lambda}{\gamma\sigma^2}$ **Constant amount of money in stock**

Solution methods: Martingale methods, convex duality, HJB-equation, …

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Note:

Log-utility and power utility are suited for financial applications, exponential utility is better suited for insurance premium calculations.

Part II: Crash-Modelling and Worst-Case Approach – 1

Hua, Wilmott (1997), K., W.(2002), K., Menkens (2005), K. Steffensen (2007), Seifried (2010), …

1. Hua & Wilmott (1997): “Number and size of crashes until a given time horizon are bounded” $\Rightarrow$ **no** probabilistic assumptions on height, number and time of crashes.

2. K. & Wilmott (2002) “Determine worst-case bounds for optimal investment”.

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For simplicity: At most one crash in [0, T] with maximum height $k^* < 1$.

$$dS(t) = S(t)\big((r+\lambda)dt + \sigma dW(t)\big)\,, \quad S(0) = s, \qquad \text{"Stock in normal times"}$$

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Consequence: Wealth process $X^{\pi}(t)$ at a (random) crash time t:

$$X^{\pi}(t-) = \big(1-\pi(t)\big)X^{\pi}(t-) + \pi(t)X^{\pi}(t-)$$

$$\Rightarrow \big(1-\pi(t)\big)X^{\pi}(t-) + \pi(t)X^{\pi}(t-)(1-k) = X^{\pi}(t-)\big(1-\pi(t)k\big) = X^{\pi}(t)$$

Part II: Crash-Modelling and Worst-Case Approach – 2

Aim: Find the best uniform **worst-case bound**, i.e. solve

$$\text{(WP)} \qquad v_1(0,x) = \sup_{\pi(.)\in A(x)} \inf_{0\le\tau\le T,\,0\le\kappa\le k^*} E\left(\frac{1}{1-\gamma}\left(X^{\pi}(T)\right)^{1-\gamma}\right)$$

i.e.

“Try to make the worst case as good as possible!”

Part II: Crash-Modelling and Worst-Case Approach – 3

Two extreme strategies (for log-utility, i.e. γ=1):

i) $\pi(t) \equiv 0$: **"Play safe"** $\Rightarrow$ Worst-case scenario: no crash (!) with worst-case bound

$$WCB_0 = E\left(\ln\left(X^0(T)\right)\right) = \ln(x) + rT$$

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ii) $\pi(t) \equiv \pi^* := \frac{\lambda}{\sigma^2}$: **"Optimal investment in the crash-free world "**

$\Rightarrow$ worst-case scenario: crash of max. height k* (at any time !) with worst-case bound

$$WCB_{\pi^*} = E\left(\ln\left(X^{\pi^*}(T)\right)\right) = \ln(x) + rT + \tfrac{1}{2}\left(\frac{\lambda}{\sigma}\right)^2 T + \ln(1 - \pi^* k^*)$$

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Insights:

- it depends on *T* which strategy performs better
- Strategy i) is **too conservative** to be good if no crash happens, strategy ii) is **too risky** to perform well if a high crash appears

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$\Rightarrow$ ***an optimal strategy should balance this out***!

Part II: Crash-Modelling and Worst-Case Approach – 4

Theorem 1 (K., Menkens (2005)) **"Optimal investment in the face of a crash"**

The unique portfolio process $\hat{\pi}(.)$ solving (WP) is given as the unique solution $\hat{\pi}(.) \in \left[0, 1/k^*\right)$ of the differential equation

$$\pi'(t) = -\frac{\gamma\sigma^2}{2k^*}\left(1-\pi(t)k^*\right)\left(\pi(t)-\pi^*(t)\right)^2, \quad \pi(T)=0 \tag{3}$$

before the crash and using the Merton strategy $\pi^*(t) = \lambda/\left(\gamma\sigma^2\right)$ **after** the crash.

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before the crash and using the Merton strategy $\pi^*(t) = \lambda/\left(\gamma\sigma^2\right)$ **after** the crash.

This strategy makes the investor **indifferent** between the worst crash happening at any time in [0, T] or happening not at all, i.e. for $t \le s \le T$ we have the indifference property

$$\textbf{(ID)} \qquad v_1(t,x) = v_0\left(t, x\left(1-\hat{\pi}(t)k^*\right)\right) = E^{t,x}\left(v_0\left(s, X^{\hat{\pi}}(s-)\left(1-\hat{\pi}(s)k^*\right)\right)\right).$$

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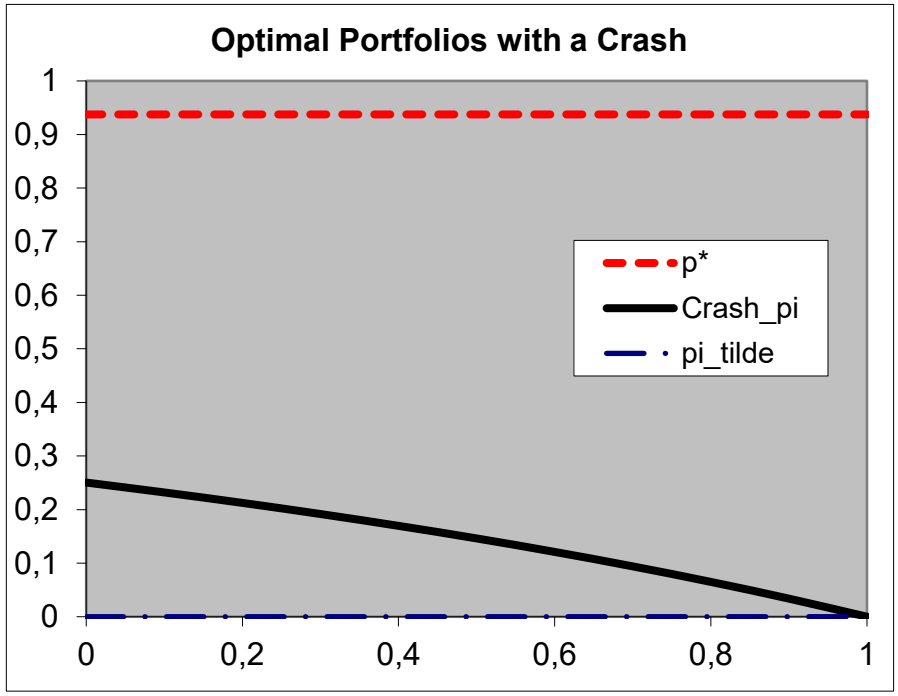
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Remark:

A strategy $\hat{\pi}(t)$ such that $M^{\hat{\pi}}(t) := v_0\left(t, X^{\hat{\pi}}(t-)\left(1-\hat{\pi}(t)k^*\right)\right)$ is a martingale is called an **indifference strategy**.

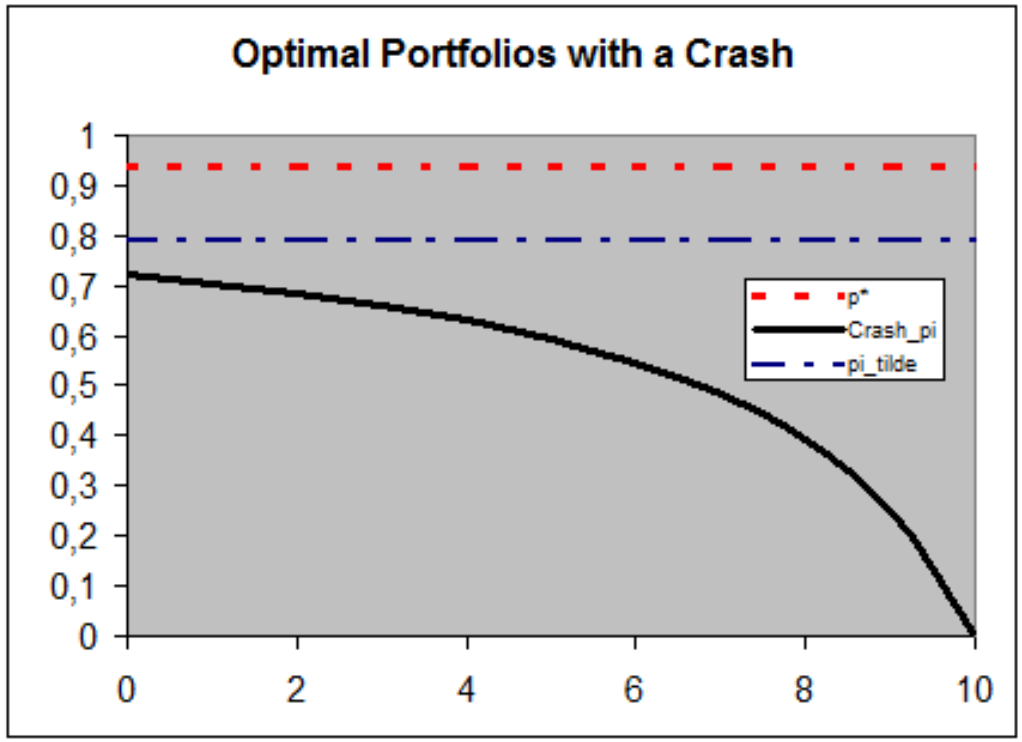
Part II: Crash-Modelling and Worst-Case Approach – 5



Example 1: $\lambda = 0.15$, $r = 0.05$, $\sigma = 0.4$, $k^* = 0{,}2$ und $T = 1$.

Optimal portfolios with and without crash, $\pi^* = 0{,}9375$.

Part II: Crash-Modelling and Worst-Case Approach – 6



Example 2: $\lambda = 0.15$, $r = 0.05$, $\sigma = 0.4$, $k^* = 0{,}2$ und $T = 10$.

Optimal portfolios with and without crash, $\pi^* = 0{,}9375$.

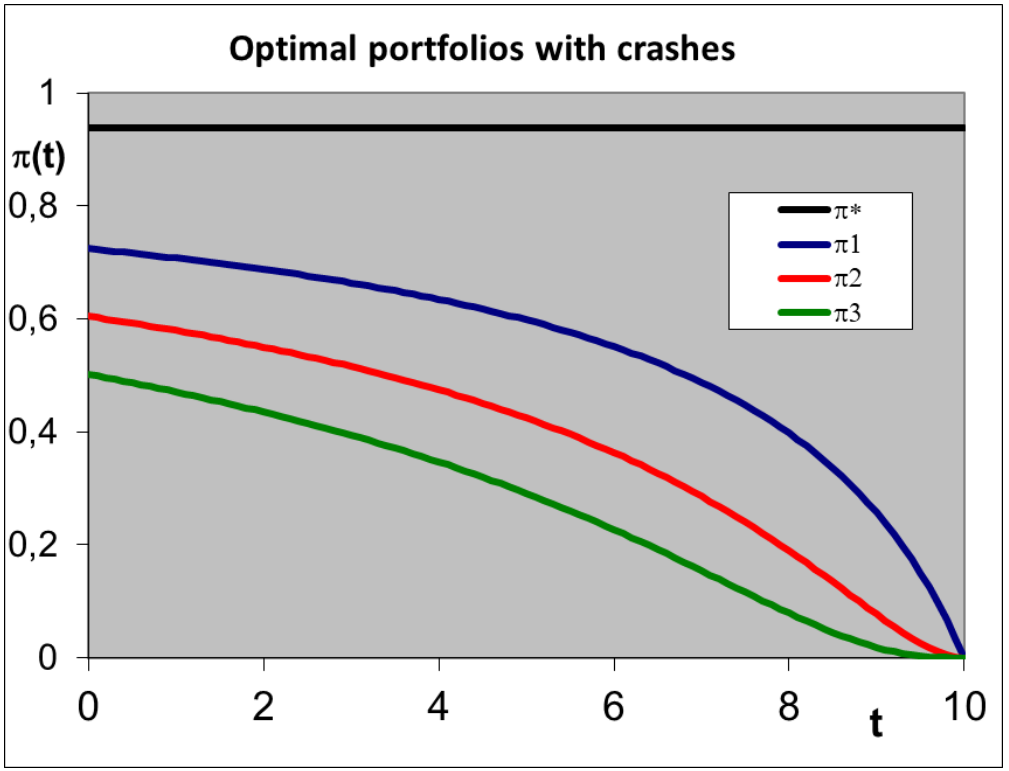
Part II: Crash-Modelling and Worst-Case Approach – 7

Modifications – *n crashes*: Similar results and methods $\rightarrow$ Use the case n =1 as a starting point for a backward induction of a system of differential equations

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Modifications – *n crashes*: Similar results and methods → Use the case n =1 as a starting point for a backward induction of a system of differential equations

Example 3: Same data as in Example 2, now with at most 3 possible crashes



Part II: Crash-Modelling and Worst-Case Approach – 8

More questions:

- What happens if we have **different types of crashes**?
- What happens if we have a **change of the market price of risk** after a crash?
- What happens if **an asset/a market participant disappears** after a crash?
- How about **consumption**?
- How about a **multi-asset setting**?
- How about **unknown market coefficients**?
- How about **more complicated price processes** in the crash-free market?
- …

Part II: Crash-Modelling and Worst-Case Approach – 9

Two fundamental results by F. Seifried (2010):

Indifference frontier ("How to improve strategies wrt. worst-case performance?")
Let $\hat{\pi}(.)$ be an indifference strategye, $\pi(.)$ an arbitrary admissible strategy and

(A) $$\tau := \inf\left\{t \geq 0 \middle| \pi(t) \geq \hat{\pi}(t)\right\} \wedge T,$$

(B) $$\tilde{\pi}(t) = \pi(t) 1_{\{[0,\tau]\}}(t) + \hat{\pi}(t) 1_{\{(\tau,T)\}}(t).$$

Then, $\tilde{\pi}(.)$ is at least as good as $\pi(.)$ with respect to the worst-case utility criterion

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Indifference-optimality principle
If there exists an indifference strategy $\hat{\pi}(.)$, such that for all other strategies $\pi(.)$ there is at least one crash time τ with

$$E\left(M^{\hat{\pi}}(\tau)\right) \geq E\left(M^{\pi}(\tau)\right),$$

then $\hat{\pi}(.)$ is worst-case optimal.

Part III: Multi-Asset Worst Case – 1

K., Leoff (2019)

Consider *d* stocks with

$$dB(t) = B(t)rdt \qquad B(0) = 1, \qquad \text{“MMA”}$$

$$dS_i(t) = S_i(t)\left((r + \lambda_i)dt + \sum_{j=1}^{i} \sigma_{ij} dW(t) \right), \quad i = 1,.., d, \qquad S_i(0) = s_i, \qquad \text{“Stock”}$$

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and crash coefficients, i.e. there is a general crash height *k* which affects each stock via

$$S_i(\tau) = S_i(\tau -)(1 - k_i) := S_i(\tau -)(1 - k\chi_i)$$

with (w.l.o.g.) **(H)** $\chi^T \pi^* > 0$.

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For a (not yet specified) indifference strategy $\hat{\pi}(.)$, introduce

$$N(t) := \chi^T \hat{\pi}(t) \qquad \text{(“sum of relative crash losses”)}$$

Part III: Multi-Asset Worst Case – 2

What is different now?

Easy to show: There exists **more than one** indifference strategy.

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Problems:

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   ⇨ Can be shown with the help of the indifference optimality principle
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Idea 1: "Minimization approach"

Find the indifference strategy that causes the smallest loss if an immediate crash happens, i.e. determine the indifference strategy $\hat{\pi}(.)$ with

(A1) $\quad \hat{\pi}(0)^T \chi = \min\left\{ \tilde{\pi}(0)^T \chi \,\middle|\, \tilde{\pi}(.) \text{ is an indifference strategy} \right\}$

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Clear(Why?): This is the (worst-case) best indifference strategy

Hard: To compute the explicit form of this worst-case optimal strategy

Part III: Multi-Asset Worst Case – 3

Idea 2: "Maximization approach"

Find the indifference strategy that performs best if no crash happens:

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Idea 2: "Maximization approach"

Find the indifference strategy that performs best if no crash happens:

Theorem 3: (Log-Utility case)

Under Ass. (H) and $\chi_i \geq 0, i = 1,\ldots,d,\ \chi \neq 0$ the optimal portfolio strategy before the crash is an indifference strategy and is given by

(A2) $$\pi_c(t) = \pi^* - \frac{\chi^T \pi^* - N^*(t)}{\chi^T \left(\sigma\sigma^T\right)^{-1} \chi} \left(\sigma\sigma^T\right)^{-1} \chi$$

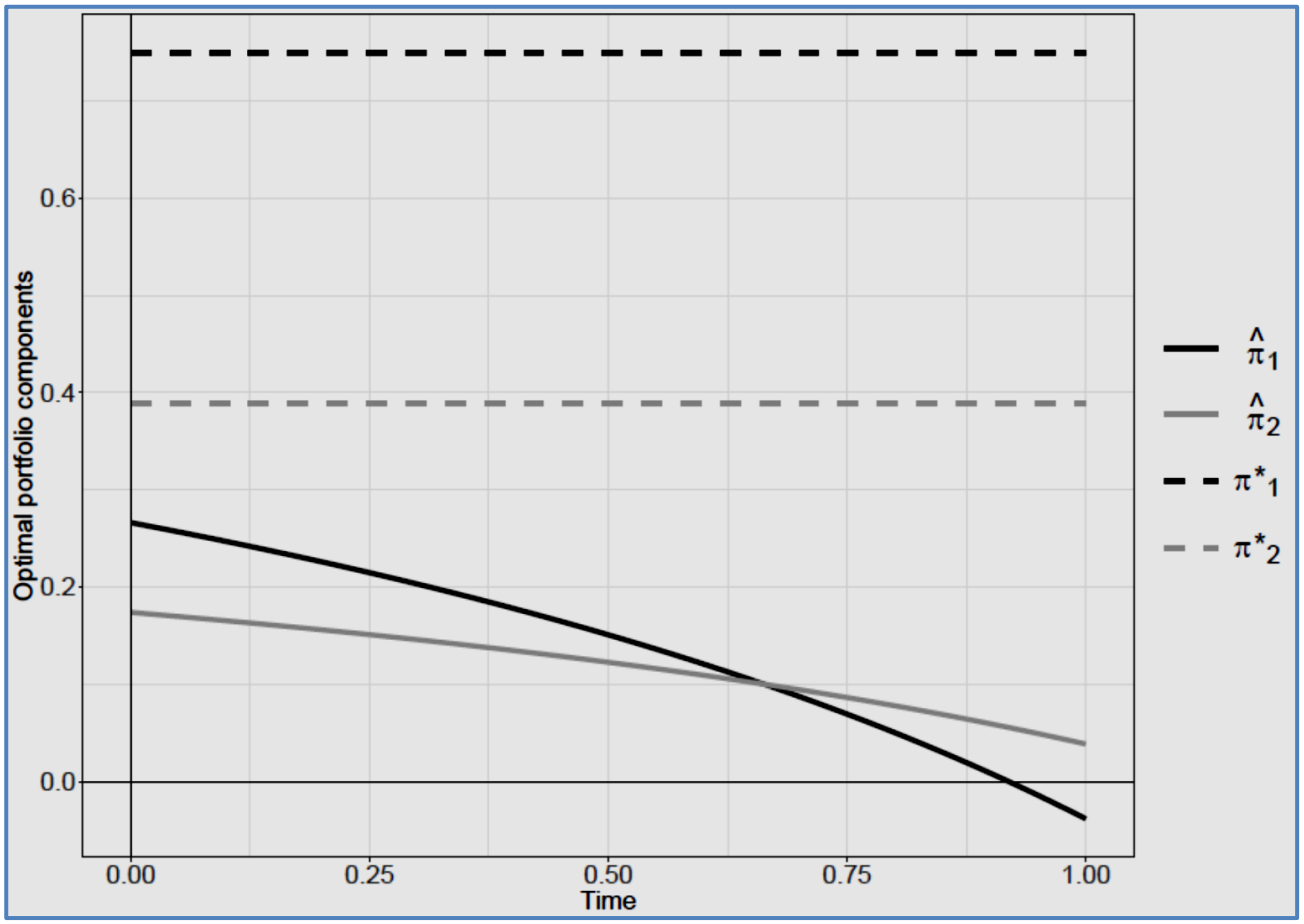
where $N^*(t)$ is the unique solution of the ODE

(A3) $$N'(t) = -\frac{1}{2k^* \chi^T \left(\sigma\sigma^T\right)^{-1} \chi} \left(1 - k^* N(t)\right)\left(\chi^T \pi^* - N(t)\right)^2,\ N(T) = 0$$

with $0 \leq N^*(t) \leq \chi^T \pi^*$.

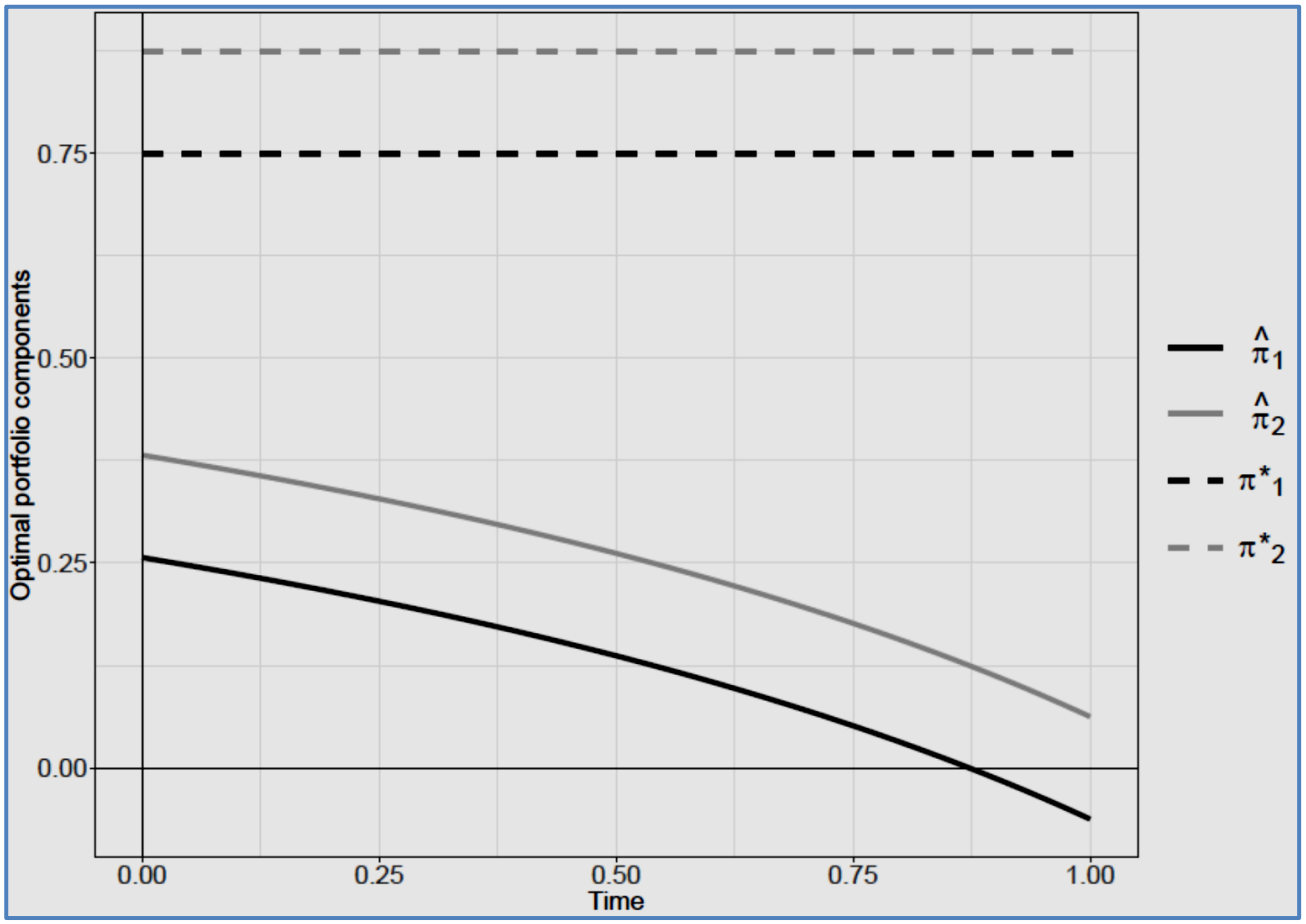
Part III: Multi-Asset Worst Case – 4

Ex.1: T=1, k*=0,025, $\chi=(1,1)^T$, r=0, b =(0,03 , 0,035), $\sigma_{11}=0,2$, $\sigma_{22}=0,3$, $\sigma_{21}=\sigma_{12}=0$



Part III: Multi-Asset Worst Case – 5

Ex.2: T=1, k*=0,025, $\chi=(1,1)^T$, r=0, b =(0,03 , 0,035), $\sigma_{11}=0,2$, $\sigma_{22}=0,2$, $\sigma_{21}=\sigma_{12}=0$



Part IV: Optimal Portfolios with Crash Scenarios and Ambiguity – 1

How about parameter uncertainty (ambiguity)?

Biagini, Pinar (2015), Garlappi et al. (2006), Hernandez-Hernandez, Schied (2006), Lin, Riedel (2021), Schied, Föllmer, Weber (2009) and many more

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Simplest example (see Lin, Riedel (2021)) Box uncertainty for drift and volatily

(A1) $dB(t) = B(t)rdt, dS(t) = S(t)[b(t)dt + \sigma(t)dW(t)]$

with $(b,\sigma) \in P_0 := \{p = (b,\sigma) | b, \sigma \text{ prog. meas.}, b(t) \in D, \sigma(t) \in V \; \forall t \in [0,T]\}$ and

(A2) $D = [\underline{b}, \bar{b}], \; 0 \notin V = [\underline{\sigma}, \bar{\sigma}], \; \bar{b} \geq r$

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Then, the optimal portfolio process under ambiguity

(A3) $$v_0(0,x) = \sup_{\pi(.) \in A(x)} \inf_{p \in P_0} E\left(\frac{1}{1-\gamma}\left(X^{\pi}(T)\right)^{1-\gamma}\right)$$

is solved by

(A4) $$\pi^*(t) = (\underline{b} - r)/(\gamma \bar{\sigma}^2)$$

Part IV: Optimal Portfolios with Crash Scenarios and Ambiguity – 2

K., Müller (2022b)

Optimal portfolio problem with stresses and ambiguity (at most one stress)

(A5) $$v_1(0,x) := \sup_{\pi \in A(x)} \inf_{(p_1,\xi,p_0)\in(P_1,\Theta,P_0)} E_{0,x}\left(\frac{1}{1-\gamma} X^{\pi}(T)^{\gamma}\right)$$

Part IV: Optimal Portfolios with Crash Scenarios and Ambiguity – 2

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Note:

a. We now have a **triple worst-case**, possibly different ambiguity in the markets before and after a crash and the usual crash worst-case approach.

b. The ambiguity before the crash results in **two possible comparisons** for the worst-case scenario, the world with **ambiguity before the crash** if no crash happens and the crash-free world with **ambiguity after the crash** if a crash happens.

Part IV: Optimal Portfolios with Crash Scenarios and Ambiguity – 2

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Keeping this in mind there are two steps:

- **Solve the portfolio problem** (A3) with P_0 and with P_1 separately
- Find a crash indifference strategy where the investor is indifferent between **no crash under P_1-ambiguity** and the worst crash leading to the **P_0-crash free ambiguity** world.

Part V: Optimal Dynamic Reinsurance with Worst-Case Reinsurer Default – 1

K., Müller (2022)

An Insurance Application motivated by a scenario from the Swiss Solvency Test (see FINMA (2020)):

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Let an insurer face a large insurance loss and at the same time the economic environment for reinsurers is bad (leading to their ratings being downgraded). A number of reinsurers default and can no longer (fully) meet their obligations.

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Let an insurer face a large insurance loss and at the same time the economic environment for reinsurers is bad (leading to their ratings being downgraded). A number of reinsurers default and can no longer (fully) meet their obligations.

As a consequence the primary insurer suffers a loss of three different kinds

- The reinsurer can no longer cover the full reinsured portion of the large loss
- The primary insurer has to buy new cover (as other reinsurers have also defaulted) and has to pay another premium for it
- Reinsurers can only partially settle the primary insurers outstanding debts from old claims

Part V: Optimal Dynamic Reinsurance with Worst-Case Reinsurer Default – 2

Mathematical setting:

- Assume a diffusion model surplus process and proportional reinsurance for the insurer.
- Let the stopping time τ be the **time of the default** of the reinsurer ($\tau=\infty$: "no default")
- $A = \{a:[0,T] \rightarrow [0,1]^T$, a(.) progress. measurable$\}$=set of **admissible risk exposures**, a_1 = risk exposure until default, a_0=risk exposure after default with a_1 having right continuous paths

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Loss of the primary insurer:

- Large loss $C > 0$ (results from the major damage that triggered the reinsurer's default)
- $(1- a_1(\tau))F$ which depends on the risk exposure due to buying new covers and only partially settled former claims

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Surplus process:

$$dR^{a,\tau}(t) = (\mu - (1 - a(t))\lambda)dt + a(t)\sigma\sigma dW(t) - L^{a,\tau}dN(t), \quad R^{a,\tau}(t) = x, \quad t\epsilon[0,T]$$

$$N(t) := 1_{\{\tau \le t\}}, \; L^{a,\tau} := 1_{\{\tau<\infty\}}(C + (1 - a_1(\tau))F)$$

with $\lambda > 0$ being the proportional premium to the reinsurer.

Part V: Optimal Dynamic Reinsurance with Worst-Case Reinsurer Default – 3

The worst-case optimization problem:

(WC_{opt}) $$\sup_{a\in A}\inf_{\tau\in\Theta} E\left(-e^{-\gamma R^{a,\tau}(T)}\right)$$

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Standard results from portfolio optimization (see also Talk 3) imply that the **optimal risk exposure after the default** is given by

(a_{opt}) $$\bar{a}_0 = \min\left\{\frac{\lambda}{\gamma\sigma^2}, 1\right\}$$

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One can then show :

Theorem 1 (**Optimal dynamic risk exposure**)

Let $\bar{a} \in A$ with $\bar{a}_0$ as defined in (a$_{opt}$) and define $\bar{a}_1$ by

i) If $\lambda < \gamma\sigma^2$, we define $\bar{a}_1(t) = \bar{a}_0 + \frac{2F}{\gamma\sigma^2(T-t)+2F/(1-\bar{a}_0)}$ for $t \in [0,T]$.

ii) If $\lambda \geq \gamma\sigma^2$, we set $\bar{a}_1(t) = 1$.

Then, the so defined $\bar{a}$ solves the optimization problem (WC$_{opt}$).

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